

Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 7

~ BookStream #002.02

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chapter 7 is called active inference and discrete time chapter 7 is the first in a pair of chapters with chapter eight on discrete and continuous time so that they're kind of like two forks of a river that we discussed in chapter four and before and describe the recipe in chapter six now seven and eight are kind of like one level deeper going from the kind of all of uh this group of animals to one level deeper into its classification scheme on the way to the specific generative model for which it's actually given in its totality but everything prior to that is about the learning about its principles and this is kind of on the trunk of the path to discrete time modeling just like chapter eight will be about continuous time modeling what would you add in okay so I think uh chapters seven and eight uh really helps to understand um in a more practical way uh how the materials from uh particularly chapters one through five applies in real time situations uh so uh even if we somehow didn't get to understand every details of chapters one through four uh by when we come to Chapters seven and eight uh I I think uh I mean some of those uncertainties uh about or about our understandings uh can be uh clarified uh in a in a at least in a practical sense so uh I believe these two chapters are really helpful um in order to consolidate our understandings from the previous chapters awesome well said so it's going to involve specifying some discrete time models 7.2 goes into perceptual processing and the general structure of the chapter is going to walk through a series of examples that building complexity where they first start with perception in 7.2 introduce decision making and then describe a few more types of motifs or cognitive structure or patterns and also check out step by step and model stream one where it's built up to in a different way so the first example is I'll let you describe it since it's Musical okay so uh yeah the first example is um the situation in which we try to uh describe uh the performance of an amateur um a musician in terms of uh how how we uh listen to the performance of an amateur musicians uh in terms of uh the the predictions we get from our anticipation of the following notes uh as opposed to

the actual notes that's being played so uh these kinds of anticipatory reaction uh listening reaction to the musician can be successfully formalized using um discrete time active inference by uh putting up uh by by putting together the matrices a uh for um the states and uh Matrix B for the transition between the states um or the transition probabilities uh which uh describe in this case describes uh the probability from going from one note to the other and uh obviously the um the actual sequence that's been played which can be described with uh The Matrix T so uh and another Point uh I wanted to point I wanted to mention is for anyone who has downloaded this chapter uh before I don't know I think about June or something I recommend re-downloading it from mit's website because uh they have corrected some of the typos that was previously present in this chapter particularly in figure 7.2 cool so this graphical model where a person is listening this is a general perceptual Bayesian framing it's specified just like with any other equations there's a lot to look into but a indicates the probability of an outcome given a state this is saying if it were all on the diagonal like an identity Matrix this is kind of a common Motif then States kind of map to themselves so in the context of uh in the context of this model A represents the mapping between the observed note and the underlying hidden true note and then B describes the transition Matrix of how those change to time D is the prior they're specified you want to describe it uh all right so in figure 7.2 or at least the incomplete version of uh figure 7.2 we see here uh well at the upper left uh part of the picture uh we see uh I mean uh the beliefs about uh about each note uh at each step at each time step and at upper right uh we somehow translate those uh beliefs into a specific numerical values uh so uh so instead of just assigning some continuous values we uh we we've simplified the situation by assigning some discrete numerical values for each note and then the lower left is supposed to show the uh the act the free energy gradients and over time uh or in other terms uh the prediction errors uh we get from um I mean comparing our predictions with the actual outcomes uh so lastly uh lower right picture uh shows uh in parallel to the upper right picture uh determines uh the values of these errors so we we can see both uh the The Continuous the initial or at least initial uh continuous assignments and values and then the further uh discretizing of the values in order to get the discrete time situation or the more tractable at this group time situations okay so it's a general passive inference task where there's priors about how states are going to change through time and then there's real data coming in so that's the kind of classical predictive coding video compression Coleman filter Bayesian setting 7.3 introduces a key Motif which is decision making and planning as inference so this is the idea of having a base graph where the variables can relate to different things there's high

composability and here the idea is that a variable is going to be proposed that we can do inference about that describes the process of decision making or policy selection so what would you say about 7.3 so 7.3 is obviously similar to what we saw in chapter 4 and if I'm not mistaken even the topology is exactly the same with that picture we saw previously so uh yeah this is the initial setup to or which acts also as a review about how these different components of General palmdp generator models needs to be described in in such situations but uh ultimately the specific case study we come across in this section is uh the attempt to model uh the behavior of uh of a mouse in a teammate so rat and a teammate so um especially uh a teammates containing an aversive stimulus in one arm and an attractive stimulus on the other so this is this can act as a kind of toy example to use uh this kind of probabilistic modeling to uh describe these situations thanks so that leads us right to figure 7.4 here's a visualization of the situation with the the rat in this case where there's a pleasant and aversive stimuli on each end of a decision point and there's also a epistemic opportunity to receive some information about the context that the animal is in and so that setting is described for both the case with white on the left block on the right and block on the left white on the right and those are shown in terms of their differences in the matrices the explicit specification of the generative model visualizations show some of the slices of the B variable which reflect different transition probabilities C represents the preferences which are expressed over the observable States D reflects the priors on the different states that need priors 7.4 what would you say about this yeah okay so in 7.4 it builds up on the previous section and adds other elements that we previously saw in chapters three and sorry two and four which is how the exact formulation for um expected free energy can be used sorry variation free energy can be used to formulate the trade-off between uh the uh I mean information uh seeking and and or at least uh between the epistemic value and information seeking so uh here it uses uh again that rad example in a bit more uh more extended and elaborate form to formulate the epistemic value of observing a q in a given a given location and figure 7.7 is a representation of this situation but another situation that's been uh let me see yeah in 7.9 another case study discussed here is uh is is the situation of uh The Psychotic eye movements and um because it is something that can be quite successful you just describe or characterize in terms of information seeking versus the epistemic value and the situation here is uh let me see yeah shown visually in figure 7.9 which clearly shows how our visual uh psychotic eye movement eye movements can be described in such a way as to as to kind of trace the trajectory uh of our eye movement among different regions of the visual space so uh and how the information We Gather from a given

region can be can affect the um I mean the subsequent trajectories of our psychotic eye movements so yeah that's basically the main premise of this section I guess nice great 7.5 what would you say about it okay so uh 7.5 again adds another dimension uh to the previous formulations and this time we get to update the generator models by learning and the so the the generative models for this situation is a bit more complicated than the previous ones because it now needs to account for a mechanism or a way to update the matrices we had before so in the previous situations uh we we didn't account for learning per se but here we directly update our general sorry that the word update can be confusing here we we get to somehow improve our generative models to accommodate for these updating accounts and yeah so uh this the situation here uh or the case of study uh uh here uh which uh somehow elucidate the way that the learning um can be a common can be accounted for uh with these models is uh again a toy example of a creature uh in a in a simple world of uh black and white tiles which kind of tries to find a a path to reach um a given destination a certain destination so it is more complicated than the situation we had for the rat example because it only had um I mean simple trajectories that needed to a Traverse but here uh the creature or the agent in this case needs to do lots of a lot more learning and information seeking and so on so all the previous elements uh is kind of combined in this example and it's a really good example to see how the different components of active inference can be connected to each other nice n76 hierarchical or deep inference first a box 7.3 interlude on structure learning boxed off topic and a lot to say but structure learning broadly refers to learning the structure about a model using the same types of methods that you might to do inference on for example a more observable sensor data reading something like that this section Works towards the idea of nested inference or or multi-scale modeling what would you say about figure 7.12 so again this situation is um I think the most complex situations of this chapter which builds up uh from the previous sections uh and this time it adds another layer to accommodate for for the inferences that happen in uh I mean different time steps so in this case we have multi-time or multi-scale inference and learning happening both at the levels of um sorry at the levels of learning and at the levels of information seeking so this this is represented in figure 7.12 which represents how uh kind of this fractal generative model can be seen as a component in um this multi-scale bigger generative or and as a kind of leaf in this bigger bigger generative model uh so it can be seen as a lower uh level inference happening at the leaf level going up to the hierarchy and influencing sorry uh collaborating on the whole process of learning and inference at the higher level so yeah I guess that's somehow summarizes this figure so if you have anything to

add that's that's great it's an example of the composability of generative models what we've uh talked about and had Toby Sinclair Smith describe as the compositional cognitive cartography and just what kinds of connectors can and can't you do and how can that Motif that the discrete time model introduces and then the rest of these features including action and learning and so on get layered in on top what can you do with that 713 gives another example do you want to say anything about it or maybe continue on uh yeah so the case study here is the example of um linguistic I mean language learning through reading so not language learning maybe it's just what happens uh sometimes yeah yeah in comprehension so what happens when reading in a anticipatory way uh the the words that that comes each after the other so uh why this kind of situation uh can be most successfully characterized with this kind of modeling because it involves a different scales of learning and comprehension both at the level of I mean reading at the level of uh somehow observing the letters and then going onto uh the words and then word groups and so on uh so uh yeah that's uh really interesting way to uh again combine all of those elements uh into a single unified model to see how those different time scales slow in the fast time scales operate together to to build this more encompassing model of more incoming generative model of the situation